AN ETALE COVER WITH A COMPACTIFICATION IS DOMINATED BY A PROPER EXTENSION AFTER BLOWUP

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3458-3467 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 3469-3482, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 36 of queue_spaces-pushouts.json (pb=656, stmt 3458-3467, proof 3469-3482).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-compactification-dominates' also exists in stacks-flat — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0F49: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3426–3435. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 087F: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 3482–3482. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let B be a quasi-compact and quasi-separated algebraic space over S. Let U be an algebraic space of finite type and separated over B. Let $V \to U$ be an étale morphism. If V has a compactification $V \subset Y$ over B, then there exists a V-admissible blowing up $Y' \to Y$ and an open $V \subset V' \subset Y'$ such that $V \to U$ extends to a proper morphism $V' \to U$.*

Proof. Consider the scheme theoretic image $Z \subset Y \times_B U$ of the "diagonal" morphism $V \to Y \times_B U$. If we replace Y by a V-admissible blowing up, then Z is replaced by the strict transform with respect to this blowing up, see Lemma 0F49. Hence by More on Morphisms of Spaces, Lemma 087F we may assume $Z \to Y$ is

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an open immersion. If $V' \subset Y$ denotes the image, then we see that the induced morphism $V' \to U$ is proper because the projection $Y \times_B U \to U$ is proper and $V' \cong Z$ is a closed subspace of $Y \times_B U$. $\square$